FutureXcel Week 2

Retail Performance Intelligence Dashboard (Dynamic Visualization Dashboard)

Analytics by Mahbub Alahi Munna

Live Dashboard: Streamlit live link

GitHub Repo: GitHub repo link

This project presents an interactive retail analytics dashboard designed to explore purchasing behavior, category performance, and sales trends using dynamic visualizations. The dashboard answers key analytical questions through multiple chart types including line, bar, histogram, scatter with trendline, heatmap, and donut chart.

How messy dataset was preprocessed:

The retail dataset was cleaned and transformed to ensure accurate analysis and visualization. Column names were standardized for consistency, and irrelevant fields such as customer and transaction identifiers were removed. The date column was converted into datetime format, and new time-based features including month, day, and weekend indicators were created.

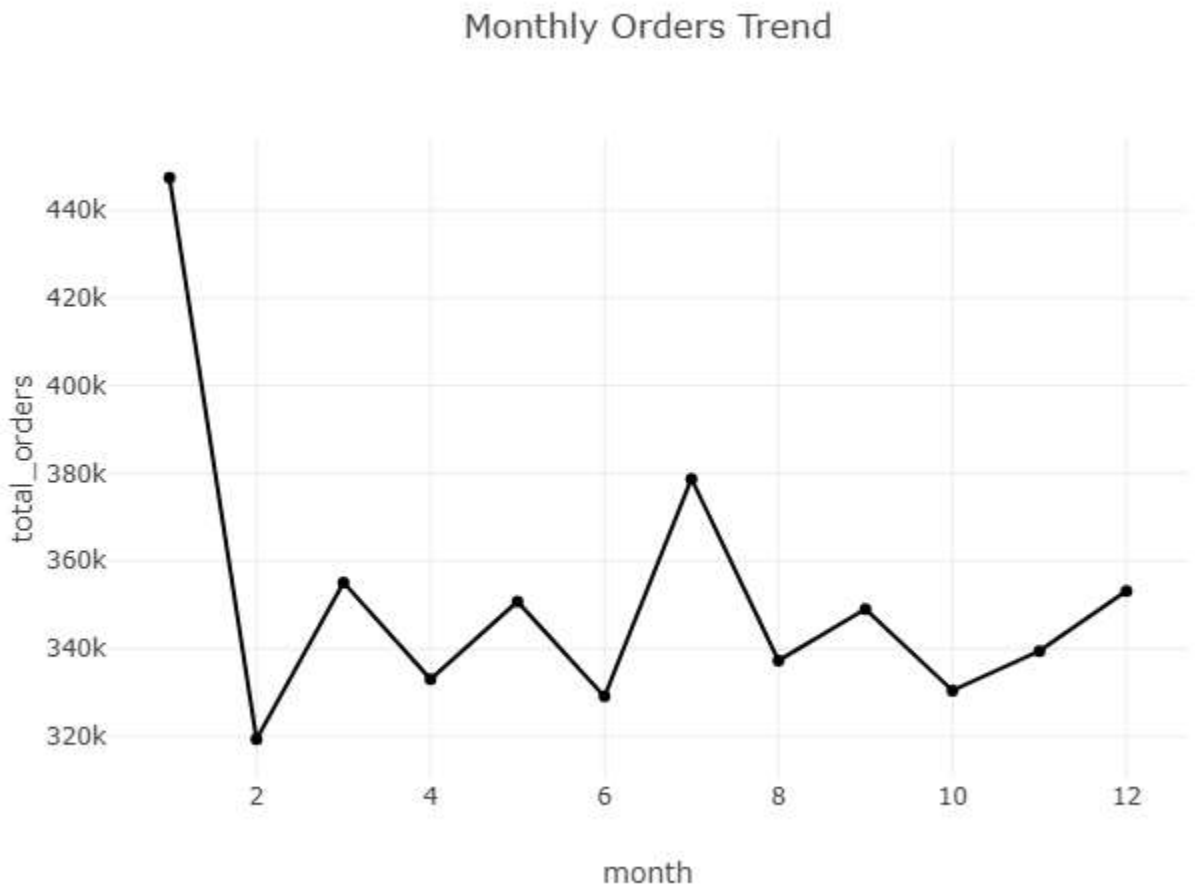
Missing values were handled using median imputation for numerical columns and “Unknown” labeling for categorical fields. Customer-level metrics such as total orders, average price, average quantity, and discount rate were engineered to capture purchasing behavior. A new feature, **cart_value**, was calculated by multiplying price per unit with quantity to support sales analysis.

Categorical variables were encoded for processing and later mapped back to readable labels to improve dashboard clarity. The final cleaned dataset was exported and used to build the interactive Streamlit analytics dashboard.

Dashboard Overview

1.Line Chart — Monthly Orders Trend :

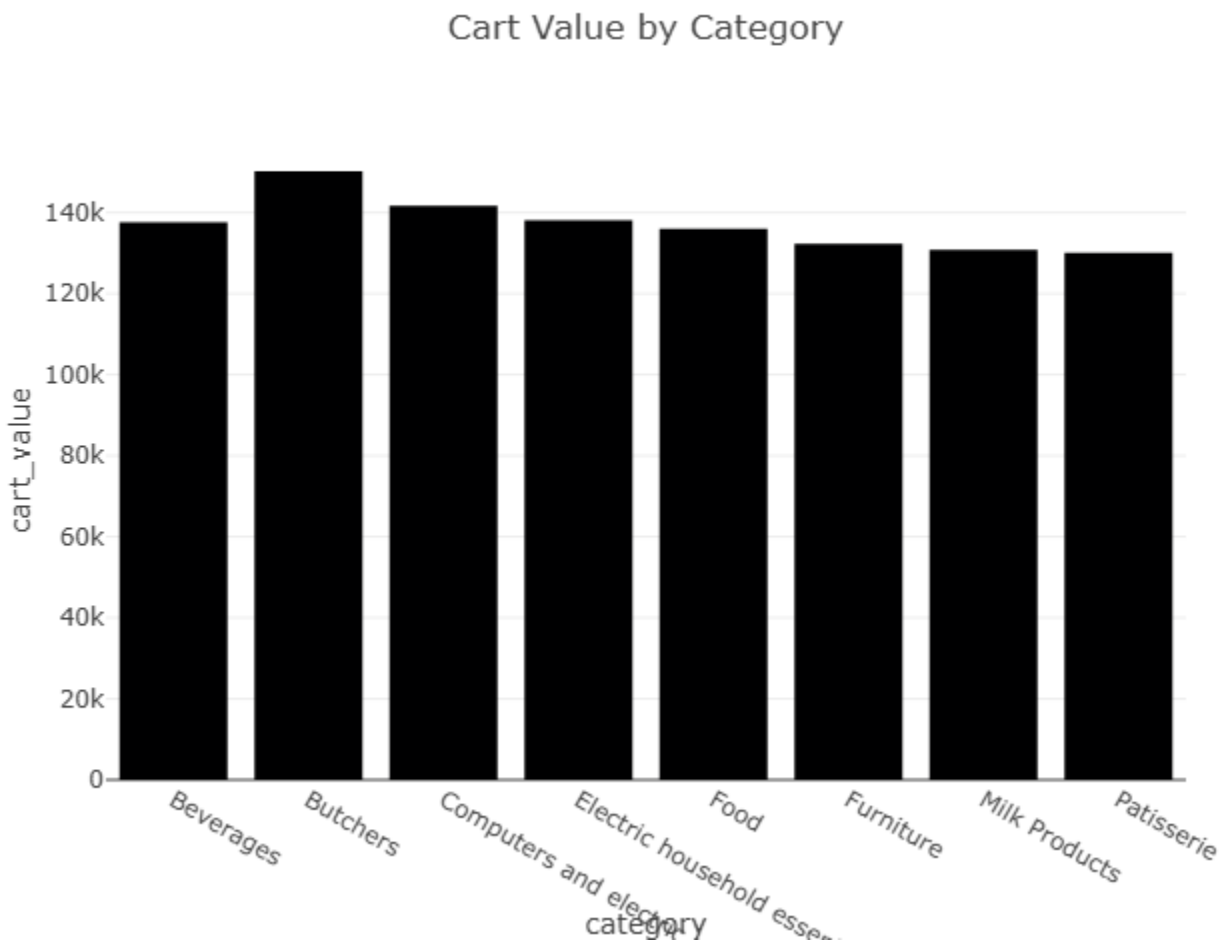
- **Orders show seasonal fluctuation patterns.**
- **Certain months indicate peak purchasing behavior.**



2. Bar Chart — Cart Value by Category :

- **Electronics and Butchers categories dominate cart value.**

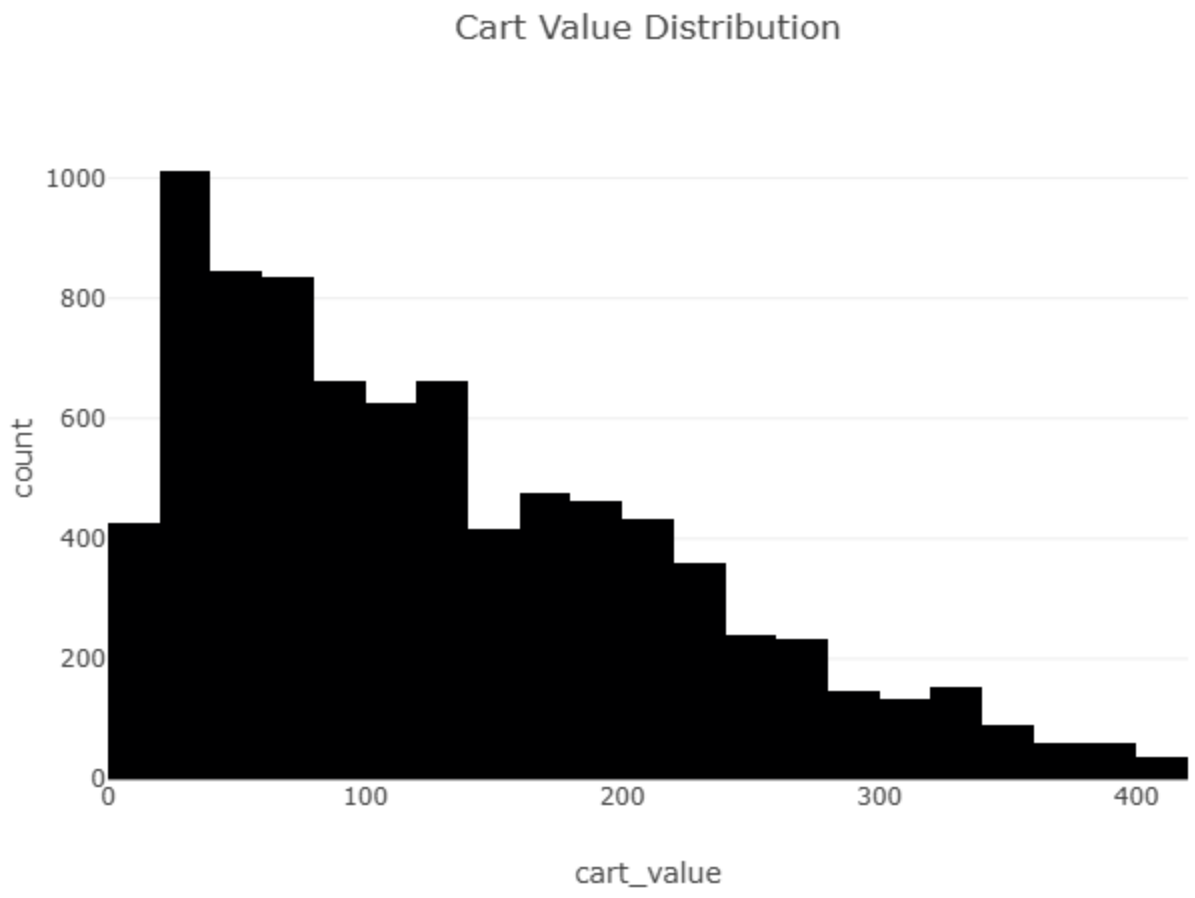
• **Lower-performing categories suggest niche demand.**



3. Histogram — Cart Value Distribution :

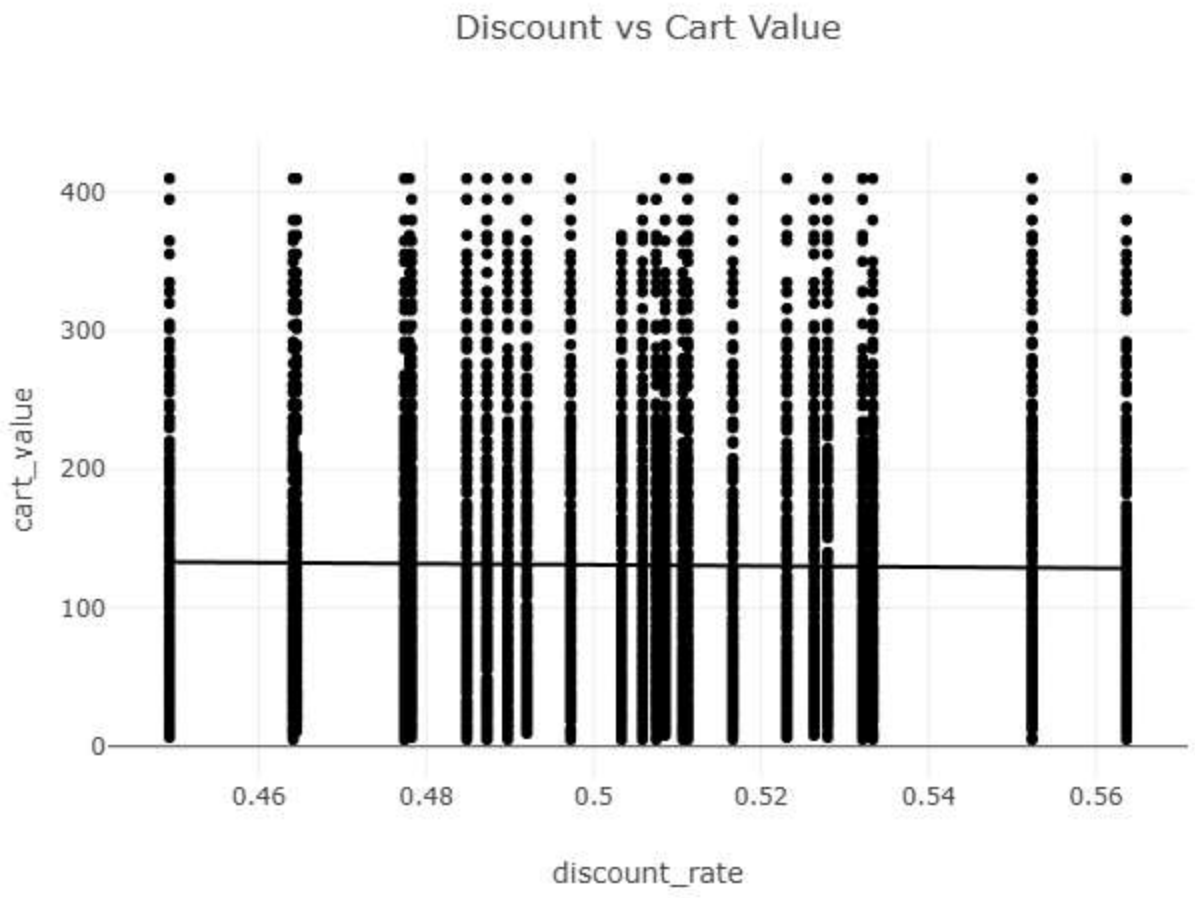
• **Most transactions fall in low-to-mid cart ranges.**

• **High-value carts are less frequent, indicating selective premium purchases.**



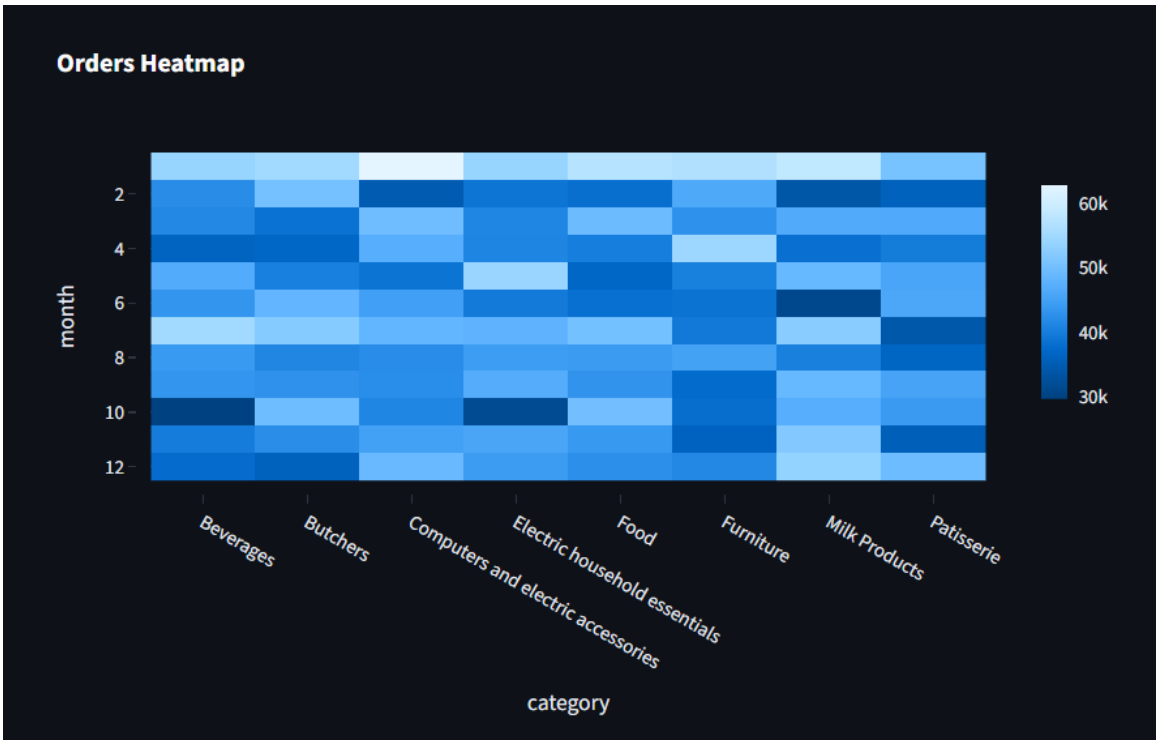
4. Scatter — Discount vs Cart Value :

- **Discount shows a moderate relationship with cart value.**
- **Higher discounts often correspond with larger purchase sizes.**



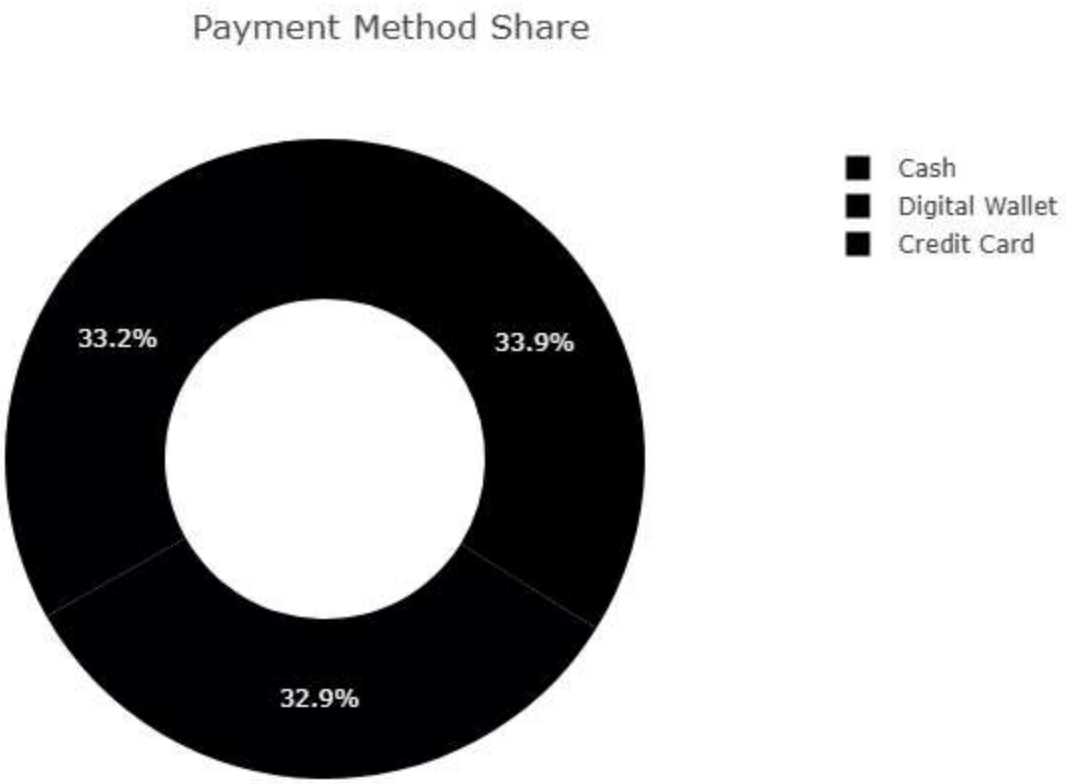
5. Heatmap — Month vs Category :

- **Seasonal peaks vary by category.**
- **Certain months drive higher order density.**



6. Donut Chart — Payment Method Share :

- **Digital payment methods lead transaction volume.**
- **Traditional payment methods still maintain steady usage.**



The Retail Performance Intelligence Dashboard demonstrates how interactive visualization can transform raw retail data into actionable insights. By combining multiple chart types and analytical storytelling, the dashboard enables clear understanding of customer behavior and operational performance.

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